

Cracking the Black Box

M5 — Deep Learning Weight Generator & Final Synthesis

Member 5 — Portfolio Replica Track

PoliMI Fintech — Business Case 3

Where M5 sits in BC3

Problem. Replicate a hidden “Monster Index”

$$r_t^{\text{target}} = 0.50 r_t^{\text{HFRX}} + 0.25 r_t^{\text{MSCI World}} + 0.25 r_t^{\text{Global Bond}}$$

using only 11 liquid Bloomberg futures (rates, equity, FX, commodity), subject to a Gaussian VaR cap of 8% (1%, 1-month).

The five tracks.

- M1 — Harness + transaction costs
- M2 — Classical linear bench (OLS / Ridge / Lasso / Huber)
- M3 — Rebalancing & constraints
- M4 — Kalman filter / state-space
- M5 — Deep learning weight generator + final synthesis

Baseline to beat. Elastic Net rolling backtest: $\rho \approx 0.72$, $\text{IR} \approx -0.26$, average gross exposure ≈ 0.22 , average VaR $\approx 2.1\%$.

The M5 contract

End-to-end weight generator. The network outputs the *weights*, not the returns:

$$w_t = f_\theta(\phi_t), \quad r_t^{\text{replica}} = w_t^\top r_t.$$

Cross-track contract (TODO.md):

- **Inputs:** $X = 11$ futures weekly returns; $y = \text{Monster Index weekly return}$.
- **Outputs:** `weights_history` $\in \mathbb{R}^{T \times 11}$, `replica_returns` $\in \mathbb{R}^T$.
- **Metrics:** $\{\text{IR}, \text{TE}, \rho, \overline{\text{GE}}, \text{VaR}, \bar{T}, \text{net_IR}, \text{net_TE}\}$.
- **Persistence:** every member commits `results/<track>.pkl`; M5 loads them into the master table.

M5 deliverables: feature engineering, MLP, training loop, rolling backtest, PCA / attention variants, consolidated comparison, slides.

ϕ_t is built from *lagged* information only (`phi.shift(1)`) so the network never sees the contemporaneous return it is asked to weight.

Components.

- Trailing compounded returns at $k \in \{4, 12, 52\}$ weeks.
- Trailing realised volatility at $k \in \{12, 52\}$ weeks (annualised $\times \sqrt{52}$).
- Regime indicators: 12-week MSCI World return, 12-week bond trend, 12-week target volatility.
- Warm start: previous Ridge/EN weight w_{t-1}^{EN} from a 52-week trailing fit ($\alpha = 10^{-3}$).

Compounded return:

$$r_t^{(k)} = \prod_{s=t-k+1}^t (1 + r_s) - 1$$

Standardisation. Train-fold μ, σ only — the network never sees val/test statistics.

Why a warm start? Inject prior structural knowledge from the EN baseline; the MLP only has to learn the residual correction.

$$\phi_t \in \mathbb{R}^{d_\phi} \rightarrow 64 \rightarrow 32 \rightarrow w_t \in \mathbb{R}^{11}$$

No softmax: long/short positions allowed. Optional *gross-exposure cap* stands in for the M3.2 / VaR feasibility projection.

```
class WeightMLP(nn.Module):
    def forward(self, phi):
        w = self.net(phi)                                # Linear -> ReLU -> Dropout -> Linear ...
        if self.gross_cap is not None:
            ge = w.abs().sum(-1, keepdim=True).clamp(min=1e-8)
            scale = torch.minimum(torch.ones_like(ge), self.gross_cap / ge)
            w = w * scale                                  # project onto {w : sum |w_j| <= L}
        return w
```

VaR projection (post-fit). If $\text{VaR}(\hat{w}_t) > 0.08$, rescale $w_t \mapsto w_t \cdot 0.08 / \text{VaR}(\hat{w}_t)$ — same rule as the EN pipeline.

Loss — tracking-error variance + turnover

$$\mathcal{L}(\theta) = \underbrace{\text{Var}_{s \in [t, t+H-1]}(w_t^\top r_s - y_s)}_{\text{TE volatility on the forward } H\text{-week window}} + \lambda \|\Delta w_t\|_1, \quad H = 12$$

- For each rebalance date t , weights w_t are scored on the realised *forward* window $[t, t + H - 1]$.
- $\lambda \in \{0, 10^{-3}, 10^{-2}\}$ ablates the turnover penalty.
- Sequential mini-batches preserve adjacency so $\|\Delta w_t\|_1$ between consecutive rows is meaningful.

Why variance, not MSE? A constant tracking bias is harmless once VaR-rescaled — what hurts is *volatility* of the tracking error. Minimising the variance directly targets the IR denominator.

Training regime — temporal split + early stopping

60 / 20 / 20 chronological split. No shuffling across the boundary — weekly financial data is non-stationary, leakage destroys the metric.

Optimiser & schedule.

- Adam, $\text{lr} = 10^{-3}$, weight decay 10^{-5} .
- Batch size 64, max 500 epochs, patience 30.
- Early stopping watches *annualised validation TE* —

$$\text{TE}^{\text{ann}} = \sqrt{52 \cdot \text{Var}(r_t^{\text{replica}} - y_t)}.$$

Reproducibility.

- seed = 42 for NumPy + PyTorch.
- Train-fold μ, σ frozen before val/test eval.
- Deterministic device fallback: CUDA if available, else CPU.

Why 60/20/20? Big enough validation tail to early-stop reliably while still leaving an out-of-sample test window the optimiser never touches.

Rolling out-of-sample backtest

The single 60/20/20 split confirms the network *can* fit. The cross-track contract requires a true rolling backtest.

`run_nn_rolling_backtest` **mirrors M1's harness:**

- Retrain on the trailing `train_window = 156` weekly samples.
- Predict $w_t = f_\theta(\phi_t)$ at every rebalance date.
- Hold weights between rebalances when $\Delta t > 1$.
- Apply the Gaussian VaR projection $\text{VaR}(\hat{w}_t) \leq 0.08$.
- Warm-start the next fit from the previous state dict — crucial for the weekly cadence.

Cadences swept. $\Delta t \in \{1, 4\}$ weeks. Same training window, loss, and VaR projection — only the retrain frequency differs.

Variants — PCA-compressed features

Replace the flat $\{4, 12, 52\}$ -week trailing-return block with the top- k principal-component scores of the trailing 52-week futures-return matrix.

- PCA is refit *causally* inside the loop: at every $t \geq 52$, fit PCA on $X[t - 52 : t]$ and store the most recent score vector.
- No future leakage; volatility, regime, and warm-start blocks unchanged.
- Default $k = 5$ components.

Why PCA? The 11 futures are heavily correlated by sector (rates, equity, commodity). A small principal subspace captures most of the cross-sectional signal and reduces d_ϕ for a leaner MLP.

Variants — Transformer over the trailing 52w matrix

A small Transformer encoder attends across $\Phi_t \in \mathbb{R}^{52 \times 11}$ and emits $w_t \in \mathbb{R}^{11}$ from the final time-step's representation.

Architecture.

- Linear projection $11 \rightarrow d_{\text{model}} = 32$.
- 2 TransformerEncoder layers, 4 heads, dropout 0.1.
- Linear head on the last token $\rightarrow w_t$, then the same gross-exposure / VaR projection.

Same harness. Loss, VaR projection, rolling cadence, train-fold standardisation are all reused — only the architecture and feature shape change. This isolates the architectural contribution.

Final consolidated comparison — master table

Synthesis step. Load each member's `results/<track>.pkl` and merge with the in-notebook NN variants.

- Each pickle ships `weights_history`, `replica_returns`, `target_returns`, `metrics`.
- Missing metrics are recomputed from the series so the master table stays homogeneous.
- Rows ranked by `net_IR` — the metric that matters once the M1 transaction-cost subtraction is applied.

Reported columns:

$$\{\text{IR}, \text{TE}, \rho, \overline{\text{GE}}, \bar{T}, \text{VaR}, \text{net_IR}, \text{net_TE}, \text{cost_drag}\}$$

Spotlight plots for the best variant: cumulative returns vs. target, rolling 26-week correlation, weights heatmap over time.

Diagnostics framework for the consolidated comparison:

- **Which model wins on IR?** Best joint signal selection + rebalance discipline.
- **Which wins on turnover?** Predictors with strong shrinkage / state-space smoothing (Kalman, NN+turnover penalty).
- **Where does the ranking flip when $\tau > \tau^*$?** Heavy-turnover NN variants lose to Kalman / Ridge once costs bite.
- **Stress windows** (2008-09 to 2009-06; 2020-02 to 2020-06): which configuration survives the crisis tail with the smallest TE / VaR breach?

What the EN baseline tells us. $\rho \approx 0.72$ at $\bar{T} \approx 0$ (long rolling window) — a stable signal beats recent adaptation. The DL bar is therefore: *lift correlation without exploding turnover.*

Recommended configuration

Architectural recipe (NN baseline).

- WeightMLP $\phi_t \rightarrow 64 \rightarrow 32 \rightarrow w_t$, dropout 0.1, gross cap $L = 2.0$.
- Features: $\{4, 12, 52\}$ -week returns + $\{12, 52\}$ -week vol + 3 regime + Ridge warm start.
- Loss: forward-window TE variance + $\lambda = 10^{-3}$ turnover penalty.
- Rebalance every $\Delta t = 4$ weeks — the cheap default; $\Delta t = 1$ retrains weekly with warm start.
- VaR projection $\text{VaR}(\hat{w}_t) \leq 0.08$, Gaussian 1% / 1-month.
- Transaction cost $\tau = 5$ bps per unit of $\bar{T}$.

When to prefer each variant.

- **MLP + warm start:** default; cheapest; respects the EN prior.
- **PCA features:** when the 11 futures are dominated by 2–3 factors and d_ϕ overfits.
- **Transformer:** when the temporal structure of the 52w window matters and compute is plentiful.

Takeaways

- **The end-to-end framing matters.** Predicting weights, not returns, gives a single coherent loss (forward-window TE variance) and lets the VaR projection live inside the model.
- **Warm-starting from EN** preserves the interpretable linear prior — the network learns the residual, not the whole map.
- **Turnover discipline is non-optional.** Without the $\lambda \|\Delta w_t\|_1$ term, NN variants over-trade and lose net IR to costs.
- **Honest evaluation.** Causal feature construction (`shift(1)`, causal PCA, train-fold standardisation), 60/20/20 split, rolling retrain — every step closed against look-ahead leakage.
- **Cross-track synthesis.** The pickle contract makes M1–M4 directly comparable in one master table; the same metric set decides the recommended configuration.

Thank you — questions?